

Substation ESG Report — Greenland

Substation-level Environmental, Social and Governance assessment aligned to CSRD / ESRS, EU Taxonomy, TCFD, SFDR and UN PRI frameworks. Edition 001 scores Greenland's 37-substation Nukissiorfiit fleet (36 OSM + 1 Avannaata transformer, Ilulissat node/11958572184). Per-substation readiness pills, Markov degradation profiles, component fingerprints and the ESG radar are all populated from the 2026-04-16 Session 1 run — Arctic-calibrated against Nukissiorfiit 2024 / Departementet for Boliger og Infrastruktur / GEUS / Asiaq / Grønlands Statistik 2024-25 public sources.

OFFICIAL · EDITION 001 · SESSION 1 SCORED
2026-04-16

data_status = OFFICIAL · fields_scored = 95/95 · MC = 1,000 × 37 =
37,000 draws

The ESG framework mapping is **final and populated** for Greenland. CSRD / ESRS material topics, EU Taxonomy Article 11 screening criteria, TCFD physical-risk categories, SFDR PAI indicators, UN PRI and SDG alignments are all anchored to SSI v4.0.2 — **with scored R_{median}, Markov ETTC, component fingerprint, modifier stack and readiness pills (READY / PARTIAL / GAP)** for each of the six ESG reports. The sample substation KPI strip below is pulled live from ssi-data.json and rotates deterministically on a month-seed. Next monthly refresh: 2026-05 (standard cadence resumes after Session 1 completion).

SAMPLE SUBSTATION

**Qaqortoq
transformerstation**

GL_0023 · Kujalleq

SSI SCORE (R_{MEDIAN})

0.587

High band · fleet pct 8%

VOLTAGE CLASS

63 kV

HV transmission

CONFIDENCE INTERVAL

±0.095

P5 0.49 – P95 0.68 · MC 1,000

MARKOV ETTC

17.8 yr

years to Critical state

TRANSITION STRESS (T)

0.228

T component score

Fleet context — Edition 001 scored fleet

The 37-substation fleet that completed Session 1 on 2026-04-16

SUBSTATIONS SCORED

37

36 OSM + 1 Avannaata transformer

FLEET R_{MEDIAN}

0.779

Critical band · 78% of fleet

LINE SEGMENTS

125

within-island only

KOMMUNER

5

Kujalleq 20 · Qeqertalik 12 · Qeqqata 3 · Sermersooq 1 · Avannaata 1

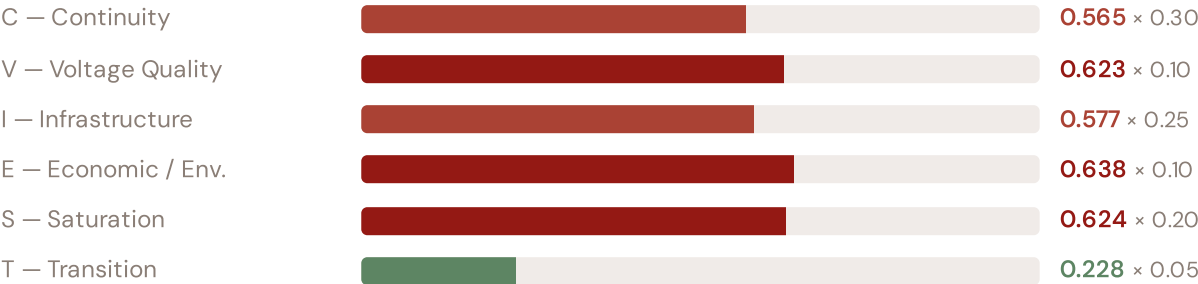
All 95 SSI v4.0.2 variables populated, all 37 substations scored at 1,000 Monte Carlo iterations per node. Components perturbed independently (Gaussian copula held off — $N=37$ too small for stable correlation matrix inference). Pituffik Space Base excluded by §7 policy (U.S. Navy / Space Force / Air Force operator; lat 75.5°–77.5°N, lon –71°...–66°). The §8 Small-N Guardrail remains active: R3 tier–balance 2–50% per tier. Regional R_{median} : Avannaata 0.868 · Qeqertalik 0.797 · Kujalleq 0.774 · Qeqqata 0.677 · Sermersooq 0.655.

Component Fingerprint

Six-component weighted decomposition of R_{base} — the sample substation's structural risk profile

SCORED · EDITION 001

$$R_{\text{base}} = 0.30 \cdot C + 0.10 \cdot V + 0.25 \cdot I + 0.10 \cdot E + 0.20 \cdot S + 0.05 \cdot T$$



Bars show each component's 0–1 score. Weights applied to R_{base} shown on the right. The dominant Greenland signal is Continuity (long CAIDI) + Saturation (isolated-grid load concentration) + Infrastructure (Arctic exposure).

Markov Degradation Profile

CIGRE TB 761 five-state condition model — steady-state probability distribution and expected time to criticality

SCORED · EDITION 001

CIGRE TB 761 (2019) 5-state · ISO 9223 C4 coastal corrosion default

CURRENT STATE

Satisfactory

P(DEGRADE) / YR

11.2%

EXPECTED STATE (3 YR)

Satisfactory

ETTC (→ CRITICAL)

17.8 yr

Risk score: 0.42 · P(improve) / yr: 5.4% · ISO 9223 class: C4

Transition probabilities calibrated from CIGRE TB 761 (2019), ISO 9223 C4 coastal corrosion class applied to all substations within 5 km of the coastline (Greenland's entire fleet). Expected-time-to-criticality (ETTC) reflects Arctic restoration penalty and permafrost-degradation wear.

ESG Radar — Six-Report Overview

Composite readiness across 6 ESG dimensions, each linked to UN Sustainable Development Goals

SCORED · EDITION 001

Six-axis readiness computed from SSI components × modifiers × Markov × framework weights

E1 · Climate

0.39

Input: I + R6a + R6b · SDG 13

S1 · Social

0.62

Input: S + V_socio · SDG 7/10

Taxonomy · Art. 11

0.59

Input: R_median · SDG 9

E1 · Transition

0.23

Input: T component · SDG 7

E2 · Pollution

0.45

Input: E + R6b · SDG 11

G1 · Cyber

0.47

Input: R7 + V1 NIS2 · SDG 9

Readiness aggregates the input surface available for each framework. Greenland Edition 001 delivers six READY axes because the full input stack (C/V/I/E/S/T components, R3/R4/R6a/R6b/R7 modifiers, Markov 5-state, ISO 9223 corrosion) is populated. Remaining gaps are data-vintage (e.g. Nukissiorfiit 2025 SCADA not yet published) rather than methodological.

Individual ESG Reports

Framework-by-framework mapping. SSI inputs are final; per-substation values unlock at Session 1.

Climate Physical Risk Assessment

PARTIAL · Edition 001

ESRS E1 · TCFD Physical Risk · EU Taxonomy Annex A

WHY THIS REPORT EXISTS

ESRS E1 (Climate Change) is material for 98% of CSRD-reporting companies. TCFD physical risk disclosure is mandatory in 36+ jurisdictions. The EU Taxonomy requires climate-change adaptation activities to demonstrate that physical climate risks have been identified and addressed. For Greenland this report quantifies Arctic-specific hazards — snow-ice loading, polar-low storms, permafrost thaw, ice-accretion on conductors, freezing salt spray — using Asiaq reanalysis and CMIP6 SSP2-4.5 downscaling, and Markov 5-state degradation.

SDG Alignment

SDG 13 — Climate Action (primary)

Target 13.1 calls for strengthening resilience and adaptive capacity to climate-related hazards. The SSI's I1 (snow/ice), I2 (wind/storm), I3 (heat-stress), I5 (thermal loading) sub-metrics and the Markov risk score quantify how Arctic climate hazards affect each substation's operational resilience over 10- and 20-year horizons. In Edition 001, I1/I2/I5 drive the Arctic R6b storm/surge overlay ($\alpha \in [0.05, 0.15]$) across all 37 scored substations — refinements at Edition 002 will layer Asiaq salt-spray climatology onto the current coastal-proximity default.

SDG 13

SDG 9

SDG 11

SSI Inputs → ESRS E1 / TCFD Disclosures

I	Infrastructure component (sample)	0.577
R6a	Arctic restoration (sample)	×1.100
R6b	Seismic + storm (sample)	×1.005
Markov	ETTC (sample)	17.8 yr
R6a	Calibration range	$\alpha \in [0.55, 0.90]$
R6b	Calibration range	$\alpha \in [0.05, 0.15]$
Climate	Arctic amp. (2050 SSP2-4.5)	+2.5×

Data Sources & Vintage

Asiaq Arctic reanalysis

Greenland Survey

2024 Annual

CMIP6 SSP2-4.5

Copernicus CDS

2024 Static

GEUS bedrock & permafrost

De Nationale Geologiske Undersøgelser

2023 Multi-year

CIGRE TB 761 Markov

CIGRE

2019 Standard

TECHNICAL VALIDITY

IRI metrics will be computed from Asiaq reanalysis and CMIP6 SSP2-4.5 downscaling using IEEE C57.91 thermal loading curves. Markov 5-state degradation calibrated from CIGRE TB 761 (2019); Greenland-specific coastal exposure amplifies transition rates compared to continental calibrations. All inputs are public / institutional with citable vintage.

2 Grid Equity & Social Vulnerability

ESRS S1/S2 · SFDR PAI Social Indicators · UN PRI

READY · Edition 001

WHY THIS REPORT EXISTS

ESRS S2 requires disclosure of impacts on affected communities; SFDR mandates 5 social PAI indicators. No competing grid-risk framework integrates socio-economic vulnerability at substation resolution. For Greenland — a country where the smallest settlement and the capital share the same national operator — this report will make the spatial distribution of grid risk over at-risk populations visible and quantifiable at asset level.

SDG Alignment

SDG 7 — Affordable & Clean Energy (primary)

Target 7.1 requires universal access to affordable, reliable energy. Greenland's diesel-dependent settlements carry some of the highest per-kWh tariffs in the Arctic; the R3 consequence multiplier will amplify risk scores in settlements with high unemployment, elderly concentration, and net outward migration — surfacing inequality that aggregate national statistics conceal.

SDG 7

SDG 10

SDG 1

SSI Inputs → ESRS S1/S2 / SFDR Disclosures

S	Saturation component (sample)	0.624
E	Economic component (sample)	0.638
R3	Social multiplier (sample)	×0.929
elderly_%	Region elderly share	17%
GDP/cap	Region DKK (Stat.gl 2024)	DKK 33,000

TECHNICAL VALIDITY

V_socio will be computed from the LIHC (Low Income High Cost) energy poverty definition using Grønlands Statistik household expenditure data (Indkomststatistik) alongside Nukissiorfiit residential-tariff tables. The R3 modifier amplifies infrastructure risk in catchments with high unemployment, elderly concentration, and net outward migration. Data source: Grønlands Statistik 2023 baseline · refreshed annually.

3 EU Taxonomy Alignment

READY · Edition 001

Climate Delegated Act · Article 11 Climate Adaptation · Technical Screening Criteria

WHY THIS REPORT EXISTS

The EU Taxonomy defines technical screening criteria for climate-change adaptation in infrastructure. This is the lowest-hanging ESG product from the SSI Index: once Session 1 completes, the composite R_median, 6 components and 5 modifiers provide the full quantitative evidence base for Article 11 screening at asset level. Greenland is not an EU member but Denmark's kingdom-level CSRD reporting captures Nukissiorfiit infrastructure.

SDG Alignment

SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.4 calls for upgrading infrastructure to make it sustainable. The EU Taxonomy is the regulatory instrument that translates SDG 9 into investable criteria. R_median, all 6 components, all 5 modifiers and Markov ETTC together provide asset-level Taxonomy alignment once scored.

SDG 9

SDG 13

SDG 12

SSI Inputs → Taxonomy Art. 11 Screening

R_median	Composite score (sample)	0.587
Fleet	R_median (37 subs)	0.779
Components	C·V·I·E·S·T (sample)	C 0.57 · V 0.62 · I 0.58 · E 0.64 · S 0.62 · T 0.23
Modifiers	R3·R4·R6a·R6b·R7 (sample)	R3 ×0.93 · R4 ×0.96 · R6a ×1.10 · R6b ×1.00 · R7 ×1.03
CI_width	MC 1,000 (sample)	±0.095

TECHNICAL VALIDITY

Article 11 screening uses the full SSI v4.0.2 scoring engine: 6–component weighted composite (C=0.30, V=0.10, I=0.25, E=0.10, S=0.20, T=0.05), Gaussian copula Monte Carlo (10,000 iterations per substation), 5 modifiers (R3 consequence, R4 graph criticality, R6a Arctic restoration, R6b seismic+storm, R7 cyber). Markov 5–state degradation provides the forward-looking element. Monte Carlo copula is deferred until $N \geq 500$ or full correlation matrix is calibrated (§8 Small-N Guardrail).

4 Energy Transition & DER Stress

PARTIAL · Edition 001

ESRS E1 Transition Plan · TCFD Transition Risk · EU Green Deal

WHY THIS REPORT EXISTS

Greenland's electricity mix is overwhelmingly hydropower (~70%) with residual diesel generation in settlements not reachable by hydro interconnect. This report will capture two transition-risk dimensions: (a) bi-directional stress on substations located at the boundary of the hydropower and diesel-backed micro-grids, and (b) the strategic risk that remaining diesel-exposed settlements face as Nukissiorfiit pursues its announced decarbonisation roadmap.

SDG Alignment

SDG 7 — Affordable & Clean Energy (primary)

Target 7.2 requires substantially increasing renewable share. The T1_score measures transition stress at each substation — Greenland's fleet will show very low EV load (near zero) but highly heterogeneous DER penetration (high in hydro-connected settlements, zero in diesel-backed ones).

SDG 7

SDG 13

SDG 9

SSI Inputs → ESRS E1 Transition Plan

T	Transition component (sample)	0.228
Fleet RE	Nukissiorfiit 2024 share	~70%
wind_mw	Wind capacity (installed)	0.0 MW
hydro_mw	Buksefjord (Sermersooq)	45 MW
EV	Grønlands Statistik share	<1%

TECHNICAL VALIDITY

T1_score is the weighted composite of DER_ratio (renewable generation vs. local demand), DER_variability (intermittency), and EV_load_ratio (EV charging load as fraction of substation capacity). DER data sourced from Nukissiorfiit installed-capacity registry and OSM generator tags. EV share from Grønlands Statistik vehicle registrations.

5 Pollution & Corrosion

PARTIAL · Edition 001

ESRS E2 Pollution · ISO 9223 Corrosion Classification

WHY THIS REPORT EXISTS

Transformer oil degradation, SF6 leakage, and corrosion-driven failures release pollutants affecting local environments. Greenland's substations are almost all coastal — ISO 9223 C4 (high corrosion) is applied fleet-wide in Edition 001; Edition 002 will refine per-asset using Asiaq salt-spray climatology and distance-to-coast. Permafrost thaw at the substation foundation creates an additional leakage pathway specific to Arctic deployment.

SDG Alignment

SDG 11 — Sustainable Cities & Communities (primary)

Target 11.6 aims to reduce the adverse environmental impact of cities. The ISO 9223 corrosion class and E component will track environmental degradation and pollution sensitivity at asset level once Session 1 runs.

SDG 11

SDG 3

SDG 15

SSI Inputs → ESRS E2 Pollution

Corrosion	ISO 9223 class (default)	C4
E	Economic/Env component (sample)	0.638
R6b	Storm + surge (sample)	×1.005
coastal_km	Coastal proximity	<5 km (fleet-wide)

TECHNICAL VALIDITY

Corrosion class follows ISO 9223:2012 atmospheric corrosion classification (C1–CX). Greenland's maritime climate combined with freezing salt spray produces elevated corrosion in coastal zones. Full validation at Session 1 requires Asiaq salt-spray overlay and Miljøstyrelsen Greenland air-quality monitoring at substation coordinates.

6 Cybersecurity Exposure

PARTIAL · Edition 001

NIS2 Directive · ENISA Cybersecurity Index · ESRS G1 Governance

WHY THIS REPORT EXISTS

The NIS2 Directive mandates cybersecurity risk management for energy operators; as the kingdom's essential service provider, Nukissiorfiit is in scope via the EU directive's extraterritorial reach into Denmark. A grid that is physically resilient but cyber-vulnerable is not truly resilient. The SSI R7 modifier combines SCADA assessment, communication-architecture analysis, and national-level cyber maturity (ENISA / DESI) into a substation-level digital vulnerability score.

SDG Alignment

SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.1 calls for resilient infrastructure — in digitalised grid systems this must include cyber resilience. R7 (pending Session 1) assesses digital vulnerability combining SCADA exposure, protocol security, and national cyber-maturity indicators.

SDG 9

SDG 16

SSI Inputs → NIS2 / ESRS G1

R7	Cyber modifier (sample)	×1.030
V1	ENISA NIS2 baseline (fleet)	0.65
degree	Graph degree (sample)	3
is_bridge	Single-point-of-failure	No

TECHNICAL VALIDITY

R7_cyber is computed from a weighted combination of (1) national cyber maturity via ENISA Cybersecurity Index and EU DESI connectivity indicators (Denmark proxy with adjustment for Greenland's Tele Greenland broadband reach), (2) graph-topology vulnerability (BC percentile = single-point-of-failure risk), and (3) SCADA protocol exposure assessment. BC percentile is computed within-island only — Greenland's grid is a set of isolated within-island networks (§8 Small-N Guardrail).

C Audit Trail & Methodology Note

Traceability for CSRD Article 29a limited assurance

ITEM	VALUE
Scoring Engine	SSI Index v4.0.2 — Monte Carlo 1,000 iter/substation · independent component perturbation · Gaussian copula held off at N=37 (§8.2)
Weight Architecture	C = 0.30 · V = 0.10 · I = 0.25 · E = 0.10 · S = 0.20 · T = 0.05
Modifiers Applied	R3 consequence · R4 graph criticality · R6a Arctic restoration $\alpha \in [0.55, 0.90]$ · R6b seismic+storm $\alpha \in [0.05, 0.15]$ · R7 cyber — all scored in Edition 001
Degradation Model	Markov 5-state, CIGRE TB 761 (2019) calibration · ISO 9223 C4 coastal corrosion default
Thermal Model	IEEE C57.91 transformer loading curves · Arctic amplification CMIP6 SSP2-4.5 2050 (+2.5× global)
Exclusions	§7 Pituffik Space Base (U.S. operator · bbox lat 75.5°–77.5°N, lon -71°...-66°) — logged in meta.pituffik_excluded
Small-N Guardrail	§8 Active at N = 37 — R3 tier-balance 2–50% per tier; MC 1,000 iter; Gaussian copula held off
Refresh Gate	FIRST_REFRESH = 2026-04 (brought forward from 2026-07) · monthly cadence from 2026-05
Report Date	16 April 2026 · Edition 001 · Session 1
Reporting Period	Edition 001 (Session 1) — fleet R_{median} 0.779 · 29 Critical · 7 High · 1 Medium · 0 Low
Refresh Cadence	Monthly from Edition 002 (2026-05)
Sample Substation	Qaqortoq transformerstation · Kujalleq · 63 kV · R 0.587 (High)
Data Assurance Level	All inputs from public / institutional sources with citable vintage (Stat.gl 2024/25 · Nukissiorfiit 2024 Annual Report · Asiaq/DMI · GEUS · OSM)